

LOGARITHMIC FUNCTIONS (Chapter 7)

LESSON 1 - Characteristics of Logarithmic Functions

Logarithmic functions are inverses of exponential functions (meaning we switch the x and the y in the equation) and have the form $y = a \log_b x$, where $b > 0$, $b \neq 1$, $a \neq 0$ and a and b are real numbers.

$$y = a^x \Leftrightarrow x = \log_a y$$

Logarithms are extremely useful when solving for an unknown exponent, especially when we cannot make the bases of the exponents the same. Their characteristics are very different than those of polynomials but are similar to those of exponential graphs.

Use your graphing calculator to graph the following functions and complete the chart below.

Note: $\log_{10} x$ is often written as $\log x$, we say “log of x to the base 10” or “log to the base 10 of x .”

For $\log x$ we say, “log x ” or “log of x .”

Logarithmic Function	Graph	x-intercept	y-intercept	$a > 0$ or $a < 0$	End Behaviour	Domain Range
$f(x) = \log_{10} x$						
$f(x) = 5 \log_{10} x$						
$f(x) = -\log_{10} x$						
$f(x) = -6 \log_{10} x$						

1. How do all of the graphs compare to exponential graphs?
2. How does the a value affect the shape of the graph?
3. Do any of the graphs cross the y -axis?

Summary

- The a value determines the _____ of the graph.
- The larger a is, the faster the function increases.
- The b value is the _____ of the log. It must always be _____ and can never equal _____.
- If $a > 0$, the graph is _____.
- If _____, the graph is decreasing.
- The domain of all logarithm graphs is _____.
- The range of all logarithm graphs is _____.

In Summary

Key Ideas

- A logarithmic function has the form $f(x) = a \log_b x$, where $b > 0$, $b \neq 1$, and $a \neq 0$, and a and b are real numbers.
- All logarithmic functions of the form $f(x) = a \log x$ and $f(x) = a \ln x$ have these characteristics:

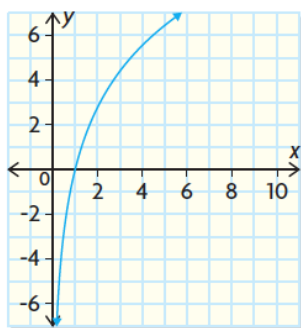
x-Intercept	1
Number of y-Intercepts	0
End Behaviour	The curve extends from quadrant IV to quadrant I or quadrant I to quadrant IV.
Domain	$\{x \mid x > 0, x \in \mathbb{R}\}$
Range	$\{y \mid y \in \mathbb{R}\}$

- All logarithmic functions of the form $f(x) = a \log x$ and $f(x) = a \ln x$ have these unique characteristics:
 - If $a > 0$, the function increases.
 - If $a < 0$, the function decreases.

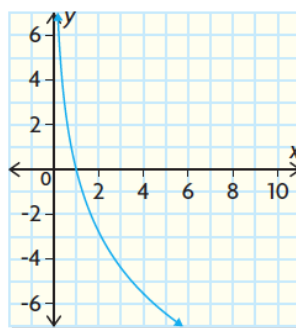
Need to Know

- The graph of a logarithmic function of the form $f(x) = a \log x$ or $f(x) = a \ln x$ will look like one of the following cases:

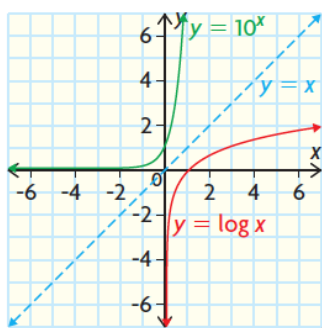
Case 1: an increasing function, where $a > 0$



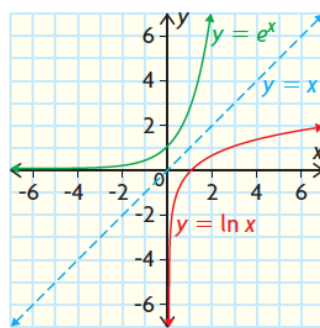
Case 2: a decreasing function, where $a < 0$



- The graph of $y = \log x$ is a reflection of the graph of $y = 10^x$ about the line $y = x$.



- The graph of $y = \ln x$ is a reflection of the graph of $y = e^x$ about the line $y = x$.



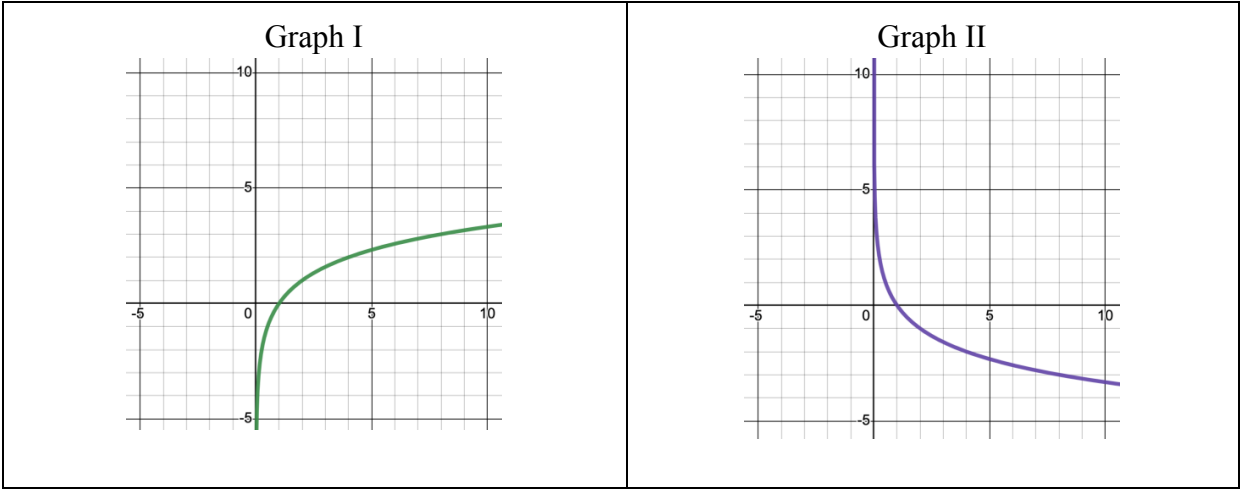
Practice

1. Answer the following questions **without using your graphing calculator.**

$f(x) = 3 \log_{10} x$	$f(x) = -7 \log_{10} x$	$f(x) = \log_e x$ *$e=2.718...$ or 2nd Fcn 
x-intercept:	x-intercept:	x-intercept:
y-intercept:	y-intercept:	y-intercept:
Increasing or decreasing?	Increasing or decreasing?	Increasing or decreasing?
Domain:	Domain:	Domain:
Range:	Range:	Range:

2. Verify your answers in #1 by graphing each function on your calculator.

3. Using the following graphs to answer the questions below.



	Graph I	Graph II
$a > 0$ or $a < 0$		
Increasing or decreasing?		
Domain		
Range		

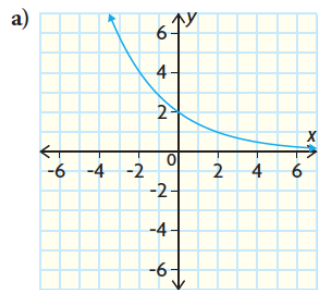
4. Match each function with the corresponding graph below. **Justify** your reasoning.

I. $y = 4.2 \log x$

II. $y = -3 \log x$

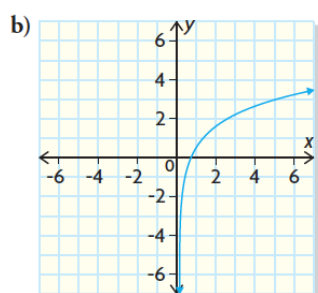
III. $y = 6^x$

IV. $y = 2(0.7)^x$



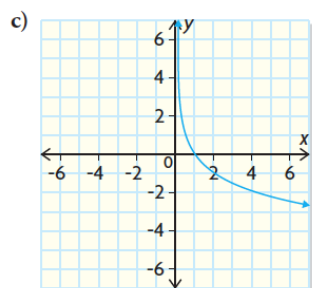
Graph: _____

Justification:



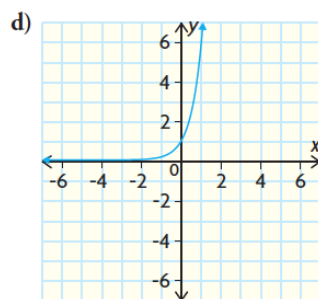
Graph: _____

Justification:



Graph: _____

Justification:



Graph: _____

Justification:

LESSON 2 - Logarithmic Regression

Just like with polynomials and exponential functions, we need to be able to determine the equation of an logarithmic function by performing a regression.

Recall: Logarithmic functions are of the form:

$$y = a \log_b x$$

where $b > 0$, $b \neq 1$, $a \neq 0$ and a and b are real numbers.

New: Logarithmic regression functions are usually of the form

$$y = a + b \ln x$$

Note: $\ln x = \log_e x$ where e is an irrational number like π .

Euler's Number, e : A constant representing an irrational number that equals 2.718....

Communication Tip

A logarithm with base e is called the natural logarithm and is written as $\ln x$. The functions $y = \log_e x$, $y = \ln x$, and $x = e^y$ are equivalent.

EXAMPLE

1. Every year Deanna determines the average height of the trees in her backyard, as shown in the table below.

Year Since Planting	Average Height (ft)
1	4.2
2	6.2
3	7.8
4	9.1
5	10.0
6	10.3
7	10.5

These data can be modelled by a logarithmic regression function in the form

$$h = a + b \ln(t)$$

where h is the average height of the trees after t years.

a) Write the logarithmic regression function of the form $y = a + b \ln(x)$, to model these data. Round values of a and b to the nearest tenth.

b) According to this regression function, the average height of these trees 12 years after Deanna planted them, to the nearest tenth of a foot, will be?

Practice

1. The length of a child, in inches, compared to time, in months, is shown in the table below.

Age (in months)	2	6	10	12	15	-	20
Length (inches)	23.9	25.4	28.3	29.9	31.2	32.3	33.4

a) Write the logarithmic regression function, $y = a + b \ln(x)$, rounded to the nearest tenth.

b) **Determine** the length of the child, to the nearest tenth of an inch, at 30 months.

c) **Determine** the age, in months, that it would take for the baby to have a length of 32.3 inches.

2. The age of a colony of bacteria in a petri dish at various population thresholds is shown in the table below.

Population	Estimated Age of Colony (hours)
20000	86.52
50000	113.05
120000	138.36

These data can be modelled by a logarithmic regression function in the form

$$y = a + b \ln(x)$$

where x is the population and y is the estimated age of the colony.

a) Write the logarithmic regression function of the form $y = a + b \ln(x)$, to model these data. Round values of a and b to the nearest tenth.

b) **Determine** the time, to the nearest hour, until the population reaches 70000.

c) **Determine** the population size after 100 hours, to the nearest whole unit.

3. Caffeine is found in coffee, tea, and soft drinks. Many people find that caffeine makes it difficult for them to sleep. The following data was collected in a study to determine how quickly the human body metabolizes caffeine. Each person started with 200 mg of caffeine in their bloodstream, and the caffeine level was measured at various times.

Caffeine in Bloodstream (mg)	Time after Ingesting (h)
200	0
168	1.0
113	5.0
90	6.5
77	8.0
50	12.0
33	14.0
23	18.0
18	20.0

a) Write the logarithmic regression function of the form $y = a + b \ln(x)$, to model these data. Round values of a and b to the nearest tenth.

b) **Determine** the time, to the nearest tenth of an hour, it takes for an average person to metabolize 50% of the caffeine in their bloodstream.

c) Steph drank a cup of coffee that contained 200 mg of caffeine at 10:00 a.m. **Determine** how much caffeine, to the nearest milligram, that will be in her bloodstream at 9:00 p.m.

d) Steph is only able to sleep if her caffeine levels are below 40 mg. If Steph usually goes to bed at 10:00 p.m., what is the latest time that she should drink a cup of coffee?

In Summary

Key Idea

- A logarithmic function may be a good model for a set of data if the points on a scatter plot form an increasing or decreasing curve, where the domain is restricted to the set of positive real numbers.

Need to Know

- The general form of the logarithmic regression model is
$$y = (\text{constant}) + (\text{multiplier}) \cdot \ln x$$
- Most graphing calculators and spreadsheets provide the equation of the logarithmic regression function in the form
$$y = a + b \ln x$$
- A logarithmic curve of best fit can be used to predict values that are not recorded or plotted. Predictions can be made by reading values from the curve of best fit on a scatter plot or by using the equation of the logarithmic regression function.

LESSON 3 - Evaluating Logarithmic Expressions & Solving Exponential Equations Using Logarithms

For every operation, there is an inverse operation that undoes the operation; For every function, there is an inverse function that undoes the function.

Recall: **Logarithmic functions** are inverses of exponential functions (meaning we switch the x and the y in the equation).

Given	$y = b^x$
Switch x and y	$x = b^y$
Isolate for y	$y = \log_b x$

Example:

$$x = 2^y \quad \rightarrow \quad y = \log_2 x$$

Notice: The **bases** in the exponential equation are the **bases** for the logarithmic equation.

“Base stays the same and the other two parts switch”

Since the logarithmic function $y = \log_b x$ is only defined for positive values of x , the logarithm of a negative number cannot be determined.



The logarithmic equation $y = \log_b x$ can be expressed in exponential form as $x = b^y$.

The exponential equation $y = b^x$ can be expressed in the logarithmic form as $\log_b y = x$.

EXAMPLE 1

Convert each of the following from exponential form to logarithmic form.

a) $4^3 = 64$

b) $2^{-3} = \frac{1}{8}$

c) $10^2 = 100$

d) $x^{2y} = 5$

e) $e^d = f$

f) $5^3 = 125$

EXAMPLE 2

Convert each of the following from logarithmic form to exponential form.

a) $\log_7 x = 4$

b) $\log_{10} 1000 = 3$

c) $m = \log_t b$

d) $\log_x a = b^4$

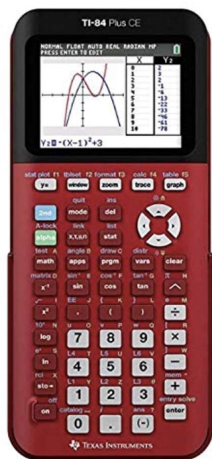
e) $5 = 4\log_b 6$

Change of Base Formula:

$$\log_b c = \frac{\log_a c}{\log_a b}$$

Typically we will make the base value, a , to be 10 so that we are working with common logarithms.

On your calculator:



TI83: Solve $y = \log_2 8$

$\log(8)/\log(2)$
3

TI84 or Higher: Solve $y = \log_5 125$

$\log_5(125)$
3

Try these:

Solve for y .

$$y = \log_3 81$$

$$y = \log_5 \sqrt{125}$$

EXAMPLE 3

Solving the following equations. Round your answers to the nearest hundredth.

a) $81 = 10^x$

b) $8.32 = 5^x$

c) $6^x = 12$

EXAMPLE 4

Solve for the variable in each of the following equations. Round your answers to the nearest hundredth.

a) $3 = \log_5 x$

b) $\log_3(3x) = 4$

c) $3\log_2 x = 4$

SUMMARY

- Exponential functions and logarithmic functions are _____ of each other.
- If we have an unknown exponent and we cannot make the bases equal, then we must use _____.
- To switch between exponential form and logarithmic form, we keep the _____ the same and _____ the other two (or look at the formula sheet).
- Our calculator uses base _____ when working with logs. To change the base, we must use the _____ formula (unless your calculator has a fancy log to multiple base capability).
- We can never take the log of a _____ number.
- $\log_a a = \underline{\hspace{1cm}}$ *this is not on the formula sheet, however it is useful to know.

Key Ideas

- The logarithmic function
 $y = \log_b x$
is equivalent to the following exponential function:
 $x = b^y$
- A logarithm is an exponent. The expression $\log_b x$ means "the exponent that must be applied to base b to give the value of x ." For example:
 $\log_2 8 = 3$
since
 $2^3 = 8$
- The value of a logarithm can be determined in one of the following ways:
 - Set the logarithmic expression equal to y , and write the equivalent exponential form. Then determine the exponent to which the base must be raised to get the required number.
 - If the base of the logarithm is 10 or e , you can use a scientific or graphing calculator.

Need to Know

- The common logarithmic function
 $y = \log x$
is equivalent to the following exponential function:
 $x = 10^y$
- The natural logarithmic function
 $y = \ln x$
is equivalent to the following exponential function:
 $x = e^y$
- The logarithm of a negative number does not exist, because a negative number cannot be written as a power of a positive base

Practice

1. Convert the following to logarithms.

a) $5^x = 36$

b) $3^{(x+2)} = 18$

c) $5 \cdot 3^{(x-1)} = 20$

2. Convert the following to exponential form.

a) $\log_5 x = 8$

b) $3 = \log_2(x - 1)$

c) $3\log_8 x = 7$

3. **Algebraically** solve the following. **Hint:** if the bases can't be made the same, switch to logarithms.

a) $2^x = 7$

b) $3(5)^x = 10$

c) $4^{3x+1} = 9$

d) $2(3)^{x+1} = 7$

e) $5\left(\frac{1}{2}\right)^{\frac{t}{6}} = 12$

f) $4(0.2)^x = 19$

4. **Solve** the following equation. $2^{4x-1} = 3^{x+1}$. *Hint: Since we cannot make the bases the same, use a graphing approach to solve.*

Extra Practice - Solving Equations

The past couple of lessons we focused on solving exponential equations, today we mix them all up so that you have to think a little more. You love it, don't you?

A couple of hints before you begin:

- Try first to make the bases the same.
- Bases can't be made the same, need to use logarithms then.
- If the unknown is inside a logarithm, switch forms to an exponential.
- We can always check/cheat by graphing and finding the point of intersection.

Solve the following **algebraically**.

$3 = 9^{2x}$	$5 = 125^{x+1}$	$10 = 3^x$
$2^{(5x)} = 13$	$8^{x-1} = 16^{3x+2}$	$2^{(x-1)} = 4^{(3x-1)}$
$4 = 16^{2x}$	$7\left(\frac{2}{3}\right)^x = 70$	$2^{(2x-5)} = 3$
$3 \cdot 2^{(x-1)} = 24$	$12 = 3\left(2\right)^{\frac{x}{14}}$	$3 \cdot 2^{(x-1)} = 70$

LESSON 4 - Laws of Logarithms

Laws of logarithms (all on formula sheet):

Multiplication Law:

$$\log_a(bc) =$$

Division Law:

$$\log_a\left(\frac{b}{c}\right) =$$

Power Law:

$$\log_a(b^c) =$$

EXAMPLE 1

Expand each of the following as a sum of logarithms.

a) $\log_5(8 \cdot 6)$

b) $\log_z(xy)$

EXAMPLE 2

Simplify each expression as a single term.

a) $\log_3 7 + \log_3 5$

b) $\log_a b + \log_a c$

c) $\log_2 5 - \log_2 3$

d) $\log_y x - \log_y z$

EXAMPLE 3

Write each expression as a single term.

a) $\log_2 A + \log_2 B - \log_2 C$

b) $3\log_2 A + 2\log_2 B - \log_2 C$

$$\text{c) } \log_2 A + \frac{1}{2} \log_2 B - \log_2 C$$

$$\text{d) } 2\log_2 A - \log_2 B - 3\log_2 C$$

EXAMPLE 4

Simplify the following *without using a calculator*.

$$\text{a) } \log_2 8$$

$$\text{b) } 3\log_4 16$$

$$\text{c) } \log_x (x^5)$$

$$\text{d) } \log_2 32 - \log_3 3$$

In Summary

Key Ideas

- The laws of logarithms are directly related to the exponent laws, since logarithms are exponents.
- The laws of logarithms can be used to simplify logarithmic expressions if all the terms have the same base.

Need to Know

- The laws of logarithms can be expressed as follows, where $b, m, n > 0$ and $b \neq 1$, and b, m , and n are real numbers:
 - Product Law of Logarithms: $\log_b mn = \log_b m + \log_b n$
For example: $\log(5 \cdot 200) = \log 5 + \log 200$
 - Quotient Law of Logarithms: $\log_b \left(\frac{m}{n}\right) = \log_b m - \log_b n$
For example: $\ln \left(\frac{3}{4}\right) = \ln 3 - \ln 4$
 - Power Law of Logarithms: $\log_b m^n = n \log_b m$
For example: $\log_2 5^3 = 3 \log_2 5$

Practice

1. Simplify each of the following, **without** your calculator.

a) $\log 10^4$

b) $\log 10^{-3}$

c) $\log_2 4^3$

d) $\log_x x^7$

e) $\log_{17} 17^4$

f) $6 \log 10$

g) $\log_2 8$

h) $3 \log_5 25$

2. Write each as a single logarithm and completely simplify.

a) $2 \log a + 5 \log b$

b) $3 \log x + 2 \log y$

c) $2 \log m - [5 \log p - \log n]$

d) $3\log 2 + 4\log 5$

e) $\log x + 4\log x - 2\log x$

f) $7\log 2 - 3\log 5 + 4\log 3$

3. Simplify $3\log_4 a - 3\log_4 b$

4. Expand $\log_4(a^5 b^7)$

5. When the expression $\log_2 5 + \log_2 3$ is simplified to the form of $\log_2 a$, the value of a is _____.

6. When the expression $4\log_2 5 - 2\log_2 2$ is simplified to the form of $\log_2 a$, the value of a is _____.

7. When $\log_b(3x) + \log_b(x^5) - \log_b(6x^2)$ is expressed as a single logarithm, the result is _____.

8. **Determine** the value of each of the following expressions and arrange them from smallest to largest.

$$\log_x(x)$$

$$\log_x(x \cdot x)$$

$$\log_x\left(\frac{x}{x^5}\right)$$

$$3\log_x(x)$$

9. When simplified $\log(2x^2) + \log(x^3) + \log\left(\frac{x}{4}\right)$ can be simplified to the form of $\log\left(\frac{Ax^B}{C}\right)$.

Find the values of A , B , and C .

10. When simplified $\log(3x^4) - [\log(9x) - \log(x^3)]$ can be simplified to the form of $\log\left(\frac{Ax^B}{C}\right)$.

Find the values of A , B , and C .

11. When simplified $2\log(4x) + \log(x) - 3\log(x^2)$ can be simplified to the form of $\log\left(\frac{A}{Bx^C}\right)$.

Find the values of A , B , and C .

LESSON 5 - Applications of Logarithmic Functions

Recall: Logarithmic functions are the **inverse** of exponential functions. Because of this we can change forms between exponential and logarithmic functions:

$$y = a^x \Leftrightarrow x = \log_a y$$

Therefore: A logarithm is equal to an exponent. The expression $\log_a y$ means “the exponent that must be applied to the base a to give the value of y .”

New:

- The fact that a logarithm is an exponent makes logarithms very useful for certain everyday-life situations where values vary greatly.
- A logarithmic scale with powers of 10 can be used to make comparisons between large and small values easier.
- Three examples of logarithmic scales are the
 - Richter scale, used to measure the magnitude of earthquakes;
 - pH scale, used to measure the acidity of solutions;
 - Decibel scale, used to measure sound levels.

Earthquakes

1. The magnitude of an earthquake, M , can be calculated using the logarithmic function

$$M = \log I$$

where I is the intensity of the earthquake measured by a seismograph.

Use this function to complete the table below.

Amplitude of vibrations on seismograph (I)	1 10^0	10 10^1	100 10^2	1000 10^3	10 000 10^4	100 000 10^5	1 000 000 10^6
Richter Scale magnitude (M)	$\log 10^0$ 0 $\log 10$ $0(1) = 0$						

- From this we see that in comparison to an earthquake with a magnitude of 4, an earthquake with a magnitude of 5 is actually _____ times more intense.
- In comparison to an earthquake with a magnitude of 2, an earthquake with a magnitude of 5 is actually _____ times more intense.
- When we write the function $M = \log I$ in exponential form, it is equivalent to _____.
- Therefore, two earthquakes with a difference of x on the Richter scale have an intensity difference of _____.
- The logarithmic Richter scale makes it easier to compare earthquakes because it uses numbers which are much more manageable.

- pH Scale**

$$pH = -\log[H^+]$$

A little review from science:



4. The pH scale is logarithmic, with a base of 10. The pH of a solution is defined by the equation

$$pH = -\log[H^+]$$

where $[H^+]$ is the concentration of hydronium ions in the solution.

- a) **Determine**, without a calculator, the pH of a solution with concentration 0.0001

(sometimes written as 10^{-4}).

- b) **Determine** the concentration of lemon juice which has $pH = 2$.

- c) **Determine** how much more acidic a solution with a pH of 1.6 is than a solution with a pH of 2.5?

5. The pH of pure water is 7. To the nearest tenth, what is the pH of a solution that is 50 times more acidic than pure water?

Loudness of Sound

The loudness of sound is measured in decibels so that the decibel measurement is $dB = 10 \log L$, where L is the relative loudness of the sound compared with the threshold of hearing.

6. The measure of a telephone dial tone is 80 dB, and the measure of an ordinary conversation is 50 dB. How many times louder is a telephone dial tone than an ordinary conversation?

7. To the nearest whole number, how many times louder is an orchestra that has a decibel level of 98 than a whisper that has a decibel level of 20?

Practice

1. Calculate the pH of a solution with each hydronium ion concentration. Round your answer to the nearest tenth.

a) 1.58×10^{-2} mol/L

b) 2.13×10^{-10} mol/L

2. **Determine** the concentration of hydronium ions in each solution
- a) Tomato juice, with a pH of 3.1 b) Milk, with a pH of 6.5
3. How many times more acidic is the acid used in car batteries than distilled water?
4. The pH level of battery acid is 0.8 and battery acid is 50.3 times more acidic than soda pop. **Determine** the pH level of soda pop , to the nearest tenth.

5. A sample of tap water has a pH of 6.3. How much more acidic is this tap water than distilled water, in terms of their hydronium ion concentrations?
6. **Determine** how many times more intense an earthquake with magnitude of 7.8 is compared to an earthquake with magnitude of 5.2.
7. The sound level of a loud indoor rock concert is 120 dB. The sound level of a quiet radio is 40 dB. How many times more intense is the rock concert than the quiet radio?

8. Diego owns a small coffee farm in Columbia. Every spring he determines the average height of the coffee bushes. The average height of the coffee bushes can be modelled by the function

$$h = 67.599 + 43.125 \ln t$$

where h is the average height of the coffee bushes in centimeters after t years.

- a) **Determine** the average height of the coffee bushes after 6 years.

- b) How many years will it take for the average height of the coffee bushes to be 160 cm?

9. Optics is the scientific study of sight and the behavior of light. As light passes through panes of glass, the glass eliminates a percentage of the light. If a single pane of glass eliminates 10% of the light that passes through it, the number of panes of glass of successive panes of glass to eliminate a particular percentage of light can be modeled by the function

$$n = -10 \ln P + 46.052$$

where n is the number of successive panes of glass and P is the percent of light allowed to pass through successive panes of glass.

- a) **Determine** the number of panes of glass needed to successfully block 50% of the light.

- b) What percent of the light is blocked by 4 panes of glass?